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Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback

Stephen Casper,[*] *MIT CSAIL,* scasper@mit.edu
Xander Davies,[*] *Harvard University*

Claudia Shi, *Columbia University*
Thomas Krendl Gilbert, *Cornell Tech*
Jérémy Scheurer, *Apollo Research*
Javier Rando, *ETH Zurich*
Rachel Freedman, *UC Berkeley*
Tomasz Korbak, *University of Sussex*
David Lindner, *ETH Zurich*
Pedro Freire, *Independent*
Tony Wang, *MIT CSAIL*
Samuel Marks, *Harvard University*
Charbel-Raphaël Segerie, *EffiSciences*
Micah Carroll, *UC Berkeley*
Andi Peng, *MIT CSAIL*
Phillip Christoffersen, *MIT CSAIL*
Mehul Damani, *MIT CSAIL*
Stewart Slocum, *MIT CSAIL*
Usman Anwar, *University of Cambridge*
Anand Siththaranjan, *UC Berkeley*
Max Nadeau, *Harvard University*
Eric J. Michaud, *MIT*
Jacob Pfau, *New York University*
Dmitrii Krasheninnikov, *University of Cambridge*
Xin Chen, *ETH Zurich*
Lauro Langosco, *University of Cambridge*
Peter Hase, *UNC Chapel Hill*

Erdem Bıyık, *University of Southern California*
Anca Dragan, *UC Berkeley*
David Krueger, *University of Cambridge*
Dorsa Sadigh, *Stanford University*
Dylan Hadfield-Menell, *MIT CSAIL*

Abstract

Reinforcement learning from human feedback (RLHF) is a technique for training AI systems to align with human goals. RLHF has emerged as the central method used to finetune state-of-the-art large language models (LLMs). Despite this popularity, there has been relatively little public work systematizing its flaws. In this paper, we (1) survey open problems and fundamental limitations of RLHF and related methods; (2) overview techniques to understand, improve, and complement RLHF in practice; and (3) propose auditing and disclosure standards to improve societal oversight of RLHF systems. Our work emphasizes the limitations of RLHF and highlights the importance of a multi-layered approach to the development of safer AI systems.

[*]Equal contribution. Correspondence to scasper@mit.edu.

1 Introduction

Reinforcement learning from human feedback (RLHF) has emerged as a prominent technique to adapt machine learning models to difficult-to-specify goals (Christiano et al., 2017; Ziegler et al., 2019; Bai et al., 2022a). In particular, RLHF is a key component of training state-of-the-art large language models (LLMs), such as OpenAI's GPT-4 (OpenAI, 2023), Anthropic's Claude (Anthropic, 2023), Google's Bard (Google, 2023), and Meta's Llama 2-Chat (Touvron et al., 2023). RLHF and similar methods allow LLMs to go beyond modeling the distribution of their training data, and adapt the distribution of text so that model outputs are rated more highly by human evaluators.

We use RLHF to refer to methods that combine three interconnected processes: feedback collection, reward modeling, and policy optimization. Figure 1 (top) illustrates this setup. The feedback process elicits evaluations of model outputs from humans. The reward modeling process uses supervised learning to train a reward model that imitates these evaluations. The policy optimization process optimizes the AI system to produce outputs that recieve favorable evaluations from the reward model. When it works well, RLHF leverages the relative ease of identifying 'good' behavior compared to demonstrations, manually-engineered reward functions, or other methods of specifying or learning rewards.

RLHF has its roots in revealed preference theory from economics. Revealed preference theory formalizes the idea that one can learn about an actor's goals from their behavior (Chambers and Echenique, 2016). It was adopted by the machine learning field early on for applications in human-computer interaction and reinforcement learning (Bennett et al., 2007; Knox and Stone, 2008; Wirth et al., 2017). The standard methodology for RLHF used today was popularized in 2017 by Christiano et al. (2017), which has played a key role in directing the attention of the deep reinforcement learning community to feedback-based methods.

RLHF has emerged as the primary strategy to finetune LLMs before deployment (OpenAI, 2023; Anthropic, 2023; Google, 2023; Touvron et al., 2023), with the goal of producing safe models aligned with human objectives. Despite this, deployed models finetuned with RLHF have revealed sensitive private information (Li et al., 2023a; El-Mhamdi et al., 2022), hallucinated untrue content (Ji et al., 2023; OpenAI, 2023; Zhang et al., 2023), spread biases that favor specific political ideologies (Santurkar et al., 2023; Perez et al., 2022b), exhibited sycophantic responses (Perez et al., 2022b), and expressed undesirable preferences (e.g., not wanting to be shut down) (Perez et al., 2022b). RLHF has also not made models robust to adversarial attacks from jailbreaking (i.e., subverting the constraints the system is normally meant to operate under) or prompt injection/extraction (Willison, 2023; Albert, 2023; Oneal, 2023; Li et al., 2023a; Wolf et al., 2023; Liu et al., 2023; Rao et al., 2023; Wei et al., 2023; Shen et al., 2023).

Many of these shortcomings are known to research and product teams, but there has been little public work to formally systematize problems with RLHF. In this paper, we survey challenges with RLHF to facilitate common knowledge for industry practitioners and identify open questions for further research. We focus primarily on applications to LLMs. We make three contributions:

1. **Concrete challenges with RLHF:** In Section 3, we taxonomize and survey problems associated with RLHF. We divide them into three primary categories: challenges with the **human feedback**, challenges with the **reward model**, and challenges with the **policy**. We also distinguish between challenges with RLHF that are more **tractable** and could be addressed within the RLHF framework using improved methodology versus **fundamental** limitations of RLHF, which require alternative approaches.[1]

2. **Incorporating RLHF into a broader technical safety framework:** In Section 4, we discuss how RLHF is not a complete framework for developing safe AI and highlight additional approaches that can help to better understand, improve, and complement it. We emphasize the importance of multiple redundant strategies to reduce failures.

3. **Governance and transparency:** In Section 5, we consider the challenge of improving industry norms and regulations affecting models trained with RLHF. Specifically, we discuss how the disclo-

[1] We use color only to highlight topics. This paper can be viewed in grayscale.

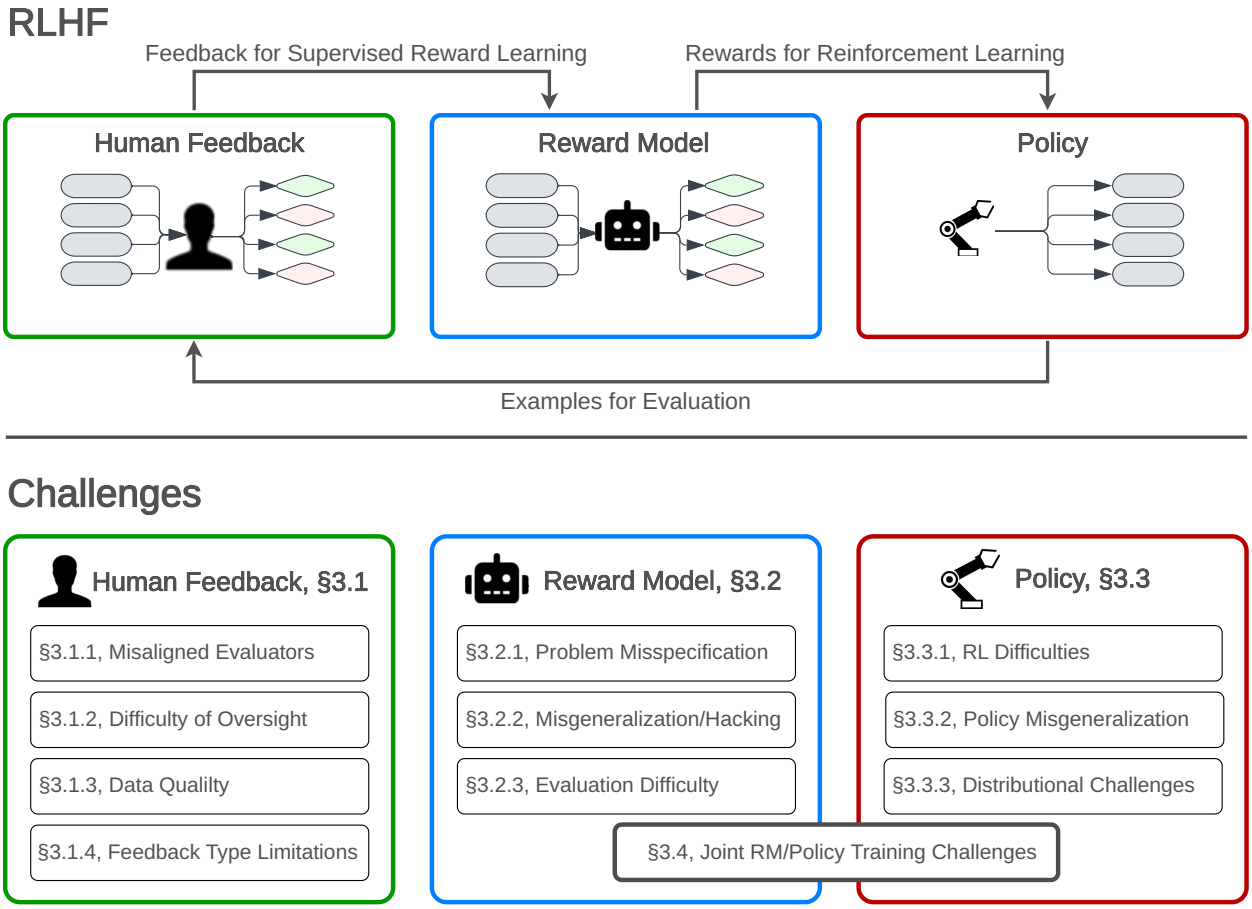

Figure 1: **(Top) Reinforcement Learning from Human Feedback.** Gray, rounded boxes correspond to outputs (e.g., text), and colored diamonds correspond to evaluations. **(Bottom) Our taxonomy for challenges with RLHF.** We divide challenges with RLHF into three main types: challenges with obtaining quality **human feedback**, challenges with learning a good **reward model**, and challenges with **policy** optimization. In the figure, each contains boxes corresponding to the subsections of Section 3.

sure of certain details by companies using RLHF to train AI systems can improve accountability and auditing.

Right now, RLHF functions both as a basic technique that can be used to study AI alignment and as a practical method to align deployed systems. Here, we focus on the possibilities and limitations of the latter. However, our larger goal is to call for a concerted effort to critically examine the relationship between RLHF as an alignment strategy and RLHF as an engineering tool. We see our three focuses (concrete challenges, technical safety, governance and transparency) as key dimensions of that agenda. Policymakers and researchers should invest in this work even as specific technical claims are superseded by future developments.

2 Background and Notation

RLHF involves three key steps: collecting human feedback, fitting a reward model, and optimizing the policy with RL. In practice, RLHF is performed iteratively by repeating these steps (or performing them synchronously). The overall procedure is illustrated in Figure 1 (top), and a specific example in which RLHF from binary preference feedback is used to finetune an LLM is depicted in Figure 2. Here, we present a simple